

Vegetable Starts — Agent-Grade Care Narrative (Priority Tagged)

Purpose: agent-readable horticultural narrative for generating care schedules and cautions. Priority tags: P1=survival-critical, P2=establishment success, P3=optimization, P4=advanced/reference. P1 statements are conservative and exception-aware.

A. Site & Light Requirements (P1)

- Most fruiting vegetables need full sun (6–8+ hours).
- Insufficient sun reduces yield and increases disease risk.

B. Temperature & Climate Constraints (P1)

- Use soil temperature to time planting.
- Cool-season benchmark: ~50°F (10°C) soil for many cool-season crops.
- Warm-season crops (tomato/pepper/basil): soil ≥60°F (16°C); tomatoes ideally 65–70°F (18–21°C).
- Protect from frost/cold nights; cold soil stalls roots and increases rot risk.

C. Seasonal Timing / Planting Window (P1)

- Warm-season transplants: plant after last frost and only when soil temps meet thresholds.
- Cool-season transplants: plant earlier when soil temps permit.
- Avoid planting into cold, waterlogged soil.
- Avoid transplanting immediately before multi-day heat wave unless irrigation/shade is available.

D. Watering & Moisture Profile (Day 0–30) (P1)

- Day 0: water in thoroughly; ensure good soil-root contact.
- First week: keep evenly moist; check daily (more often in heat/wind).
- After week 1: water deeply when top 1–2 inches are dry; avoid chronic saturation.
- Water at base; avoid wetting foliage late in day.

E. Soil Requirements (P2)

- Well-drained, fertile soil; incorporate compost broadly.
- Raised beds improve drainage and soil warming.

F. Planting & Installation Rules (P1)

- Correct depth varies by crop (tomatoes can be deeper; many others at pot depth).
- Firm soil around roots; eliminate air gaps.
- Mulch after soil has warmed for warm-season crops to avoid keeping soil cold.

G. Fertilization Guidance (P2)

- A dilute starter fertilizer is commonly used for transplants; apply only at label-safe dilution to avoid burn.
- Do not over-fertilize; burn damage is common.
- Delay stronger feeding until plants resume active growth.

H. Pest & Disease Risk (Early Window) (P2)

- Inspect for cutworms/aphids; use collars for cutworms where common.
- Cold, wet conditions increase rot risk; prioritize drainage and soil warmth.

I. Weed Competition Sensitivity (P2)

- Keep weed-free early; weeds compete and harbor pests.

J. Weather Exception Handling (P1)

- Frost: cover overnight; remove in morning.
- Heat wave: provide afternoon shade; increase watering; avoid fertilizing during heat stress.
- Heavy rain: ensure drainage; avoid working wet soil (compaction).

K. Growth & Establishment Signals (P2)

- Normal: slight wilt on planting day; recovery within hours after watering.
- Concerning: persistent wilt with moist soil, stem collapse, fast-spreading discoloration.

L. Safety & Customer Guardrails (P1)

- Do not plant warm-season crops into cold soil.
- Do not apply fertilizer to dry soil; water first.

M. Location-Based Modifiers (P2)

- Short seasons: use soil thermometer and row covers; time mulch for warming.
- Hot/humid: prioritize airflow and early-day watering.

N. Lifecycle Transition (Post 30 Days) (P4)

- Move to crop-specific schedules (staking, pruning, feeding) after establishment.